A Homogeneous Compensation Mechanism for Indirect Evolutionary Rescue in a Roy-Style Predator–Prey Model

Mathbio research notes

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Abstract

Indirect evolutionary rescue can occur when evolutionary change in one species prevents extinction of another. We study this possibility in a Roy-style predator–prey model with evolving prey defense, comparing a well-mixed eco-evolutionary ODE with a spatial PDE using the same local reaction terms. The supported mechanism is a homogeneous compensation branch: as predator mortality stress increases, the equilibrium prey defense frequency decreases, increasing predator conversion opportunity and preserving a positive predator equilibrium over an interior stress interval. For the linear defense trade-off ODE, the branch follows from explicit feasibility conditions, and its local stability is supported by Routh–Hurwitz inequalities for the cubic characteristic polynomial at the tested stresses. In the spatial PDE, the same homogeneous branch is linearly stable to tested Neumann spatial modes, and targeted finite-amplitude non-homogeneous perturbations do not produce persistent spatial-pattern-mediated rescue in the tested setup. A controlled nonlinear trade-off extension using concave, convex, and mixed endpoint-preserving shape functions supports compensation beyond the linear case, while showing parameter and shape sensitivity. The current evidence therefore supports homogeneous reaction-level compensation, not spatial-pattern-mediated rescue, for the tested parameterization.

1 Introduction

Evolutionary rescue describes persistence after environmental deterioration through adaptive change [Gomulkiewicz and Holt, 1995, Bell and Gonzalez, 2009, Bell, 2017]. In predator–prey systems, rescue can also be indirect: evolution in prey may alter predator persistence. In the setting studied here, prey defense reduces predation but carries ecological costs. When predator mortality stress increases, predator pressure declines, selection can shift prey toward lower defense, and this lower-defense prey state can increase predator conversion opportunity. This feedback is the qualitative mechanism of indirect evolutionary rescue discussed by Yamamichi and Miner [2015].

Prey defense evolution is a natural target for this question because the defense trait changes both prey ecological performance and predator energetic gain. Rapid eco-evolutionary feedbacks in predator–prey systems can alter ecological dynamics on short timescales [Yoshida et al., 2003]. The exact form of the defense trade-off is also important: nonlinear or asymmetric trade-offs can change predator–prey eco-evolutionary dynamics [Kasada et al., 2014].

Spatial structure was initially considered because local dispersal and spatial heterogeneity can change predator–prey overlap, delay extinction, and alter basin entry. A scalar persistence threshold is an attractive first diagnostic: one compares the mortality stress at which predators persist or fail in the ODE and PDE. That threshold framing can fail, however, when long transients, continuation dependence, or multiple reachable outcomes occur at the same stress. In that case, the correct object is not a single threshold but the mechanism organizing the reachable basins.

The final mechanism supported by the present analysis is not spatial-pattern-mediated rescue. Fixed-defense spatial patterning did not produce robust predator rescue. The well-mixed eco-evolutionary ODE did support indirect evolutionary rescue. Subsequent ODE–PDE comparison showed that the basin structure observed in the PDE is primarily inherited from homogeneous reaction dynamics rather than generated by persistent spatial patterning. The important mechanism is therefore a homogeneous compensation branch, with spatial analyses used to test whether the PDE destabilizes or qualitatively alters that branch.

We ask whether the rescue mechanism is controlled by spatial patterning or by homogeneous eco-evolutionary reaction dynamics. We then derive the compensation branch responsible for rescue, test its local stability, test whether the spatial PDE destabilizes it, and examine whether the mechanism persists under controlled nonlinear trade-off shapes.

2 Model

Let n be total prey density, w predator density, q prey defense frequency, and

$$z = \kappa^{-1} - n - w$$

be free space. For the linear trade-off model,

$$r(q) = r_u(1-q) + r_v q, \quad a(q) = a_u(1-q) + a_v q, \quad b(q) = b_u(1-q) + b_v q.$$

The well-mixed ODE is

$$\frac{dn}{dt} = n[r(q)z - \xi - a(q)w], \tag{1}$$

$$\frac{dw}{dt} = w[b(q)nz - (m+s) - \mu w], \tag{2}$$

$$\frac{dq}{dt} = \nu q(1-q)[(r_v - r_u)z - (a_v - a_u)w]. \tag{3}$$

Here s is additional predator mortality stress. The current parameterization uses

$$r_u = 1, \quad r_v = 0.65, \quad a_u = 1, \quad a_v = 0.35,$$

$$b_u = 0.08, \quad b_v = 0.02, \quad \kappa = 0.15, \quad \xi = 0.55, \quad m = 0.1, \quad \mu = 0.2, \quad \nu = 0.05.$$

The spatial PDE uses the same local reaction terms:

$$\partial_t n = D_n \nabla^2 n + n[r(q)z - \xi - a(q)w], \tag{4}$$

$$\partial_t w = D_w \nabla^2 w + w[b(q)nz - (m+s) - \mu w], \tag{5}$$

$$\partial_t q = D_q \nabla^2 q + \nu q(1-q)[(r_v - r_u)z - (a_v - a_u)w]. \tag{6}$$

The PDE tests use zero-flux Neumann boundaries. The current diffusion coefficients are $D_n = 0.01$, $D_w = 0.01$, and $D_q = 0.005$.

For the controlled nonlinear trade-off extension, endpoint values are preserved but linear interpolation is replaced by shape functions:

$$r(q) = r_u + (r_v - r_u) f_r(q),$$

$$a(q) = a_u + (a_v - a_u) f_a(q),$$

$$b(q) = b_u + (b_v - b_u) f_b(q),$$

with $f_\gamma(q) = q^\gamma$. This nonlinear form is used only for controlled extension tests, not to replace the baseline linear model.

3 Homogeneous compensation branch

Define

$$\Delta r = r_v - r_u, \quad \Delta a = a_v - a_u, \quad \Delta b = b_v - b_u.$$

For an interior equilibrium with $n > 0$, $w > 0$, and $0 < q < 1$, the nontrivial equilibrium conditions are

$$r(q)z - \xi - a(q)w = 0,$$

$$b(q)nz - (m + s) - \mu w = 0,$$

$$\Delta r z - \Delta a w = 0.$$

If

$$c = \frac{\Delta r}{\Delta a},$$

then the selection-gradient condition gives

$$w^* = cz^*.$$

Because the trade-offs are linear, $r(q) - ca(q) = r_u - ca_u$, so the prey-balance equation gives

$$z^* = \frac{\xi}{r_u - ca_u},$$

$$w^* = cz^*,$$

$$n^* = \kappa^{-1} - z^* - w^*.$$

The predator-balance equation then determines the defense frequency required at stress s:

$$q^*(s) = \frac{(m + s + \mu w^*)/(n^* z^*) - b_u}{b_v - b_u}.$$

This branch explains rescue as compensation: increasing predator mortality stress is offset by decreasing equilibrium prey defense frequency, which increases predator conversion opportunity and keeps a positive predator equilibrium feasible. In the current parameterization, the branch has approximately $n^* \approx 4.8333$ and $w^* \approx 0.6417$, while $q^*(s)$ decreases with stress.

4 Existence and stress interval

The interior branch exists only under explicit feasibility conditions:

$$c > 0,$$

$$r_u - ca_u > 0,$$

$$z^* > 0, \quad w^* > 0, \quad n^* > 0,$$

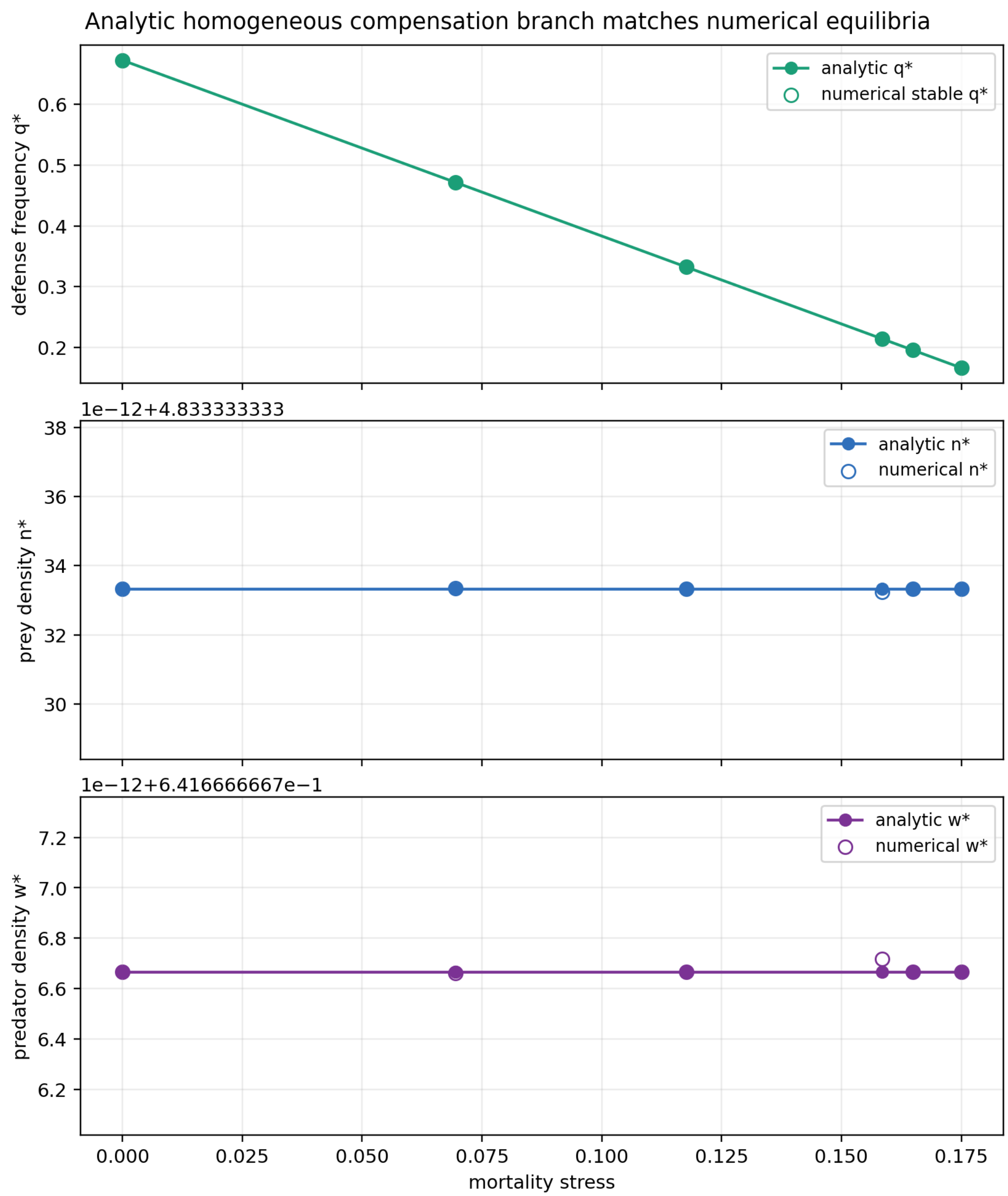

Figure 1: Analytic compensation branch for the current linear trade-off parameterization, compared with numerical stable persistent equilibria. The branch keeps n^* and w^* approximately fixed while $q^*(s)$ declines with predator mortality stress.

$$0 < q^*(s) < 1.$$

Solving the branch equation for stress gives

$$s(q) = n^* z^* [b_u + \Delta b q] - m - \mu w^*.$$

Therefore the open stress interval for an interior branch is the interval between the values at $q = 0$ and $q = 1$. In the current parameterization this is approximately

$$s \in (-0.1131, 0.2324).$$

The target stresses used in the mechanism tests lie inside this interval. These conditions make the branch a conditional mathematical statement for the linear trade-off ODE rather than a purely numerical observation.

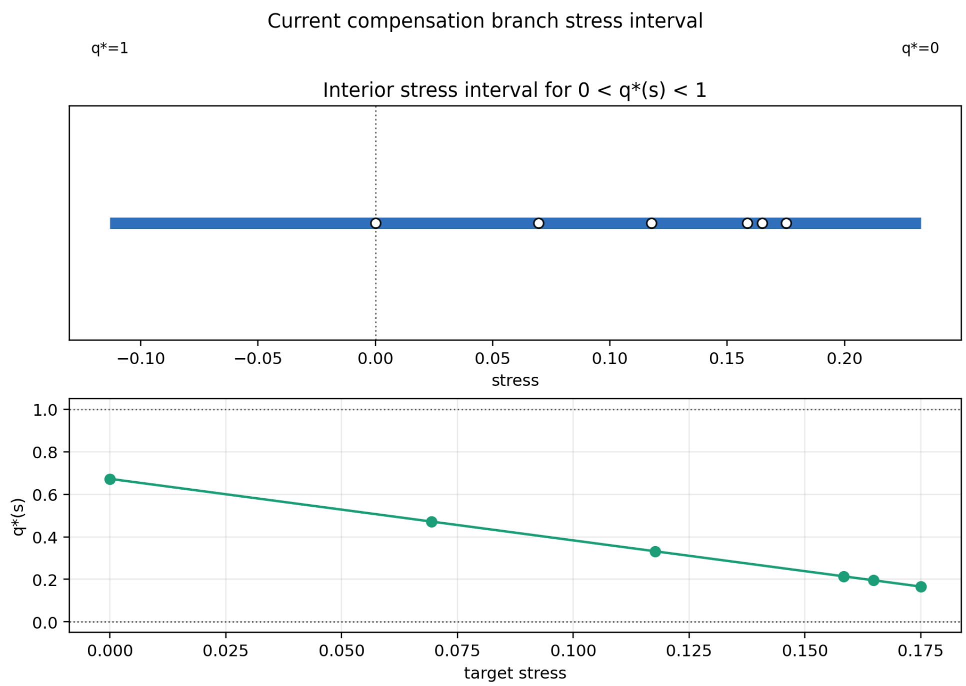

Figure 2: Current interior stress interval for the linear compensation branch. The branch is feasible where $0 < q^*(s) < 1$; the target stresses fall inside the interval.

5 Local stability of the compensation branch

Local stability of the homogeneous branch was evaluated with the analytic Jacobian of the ODE. For a three-dimensional system, the characteristic polynomial can be written

$$\lambda^3 + A_1 \lambda^2 + A_2 \lambda + A_3 = 0.$$

The cubic Routh–Hurwitz conditions for local stability are

$$A_1 > 0, \quad A_2 > 0, \quad A_3 > 0, \quad A_1 A_2 > A_3.$$

For the current parameterization, these inequalities hold at all target stresses tested along the branch and agree with direct eigenvalue stability. The smallest recorded Routh–Hurwitz margin $A_1A_2 - A_3$ across the target stresses was positive. This supports local stability of the homogeneous compensation branch in the tested stress interval.

This is a local stability result. It does not prove global attraction or determine all basin boundaries.

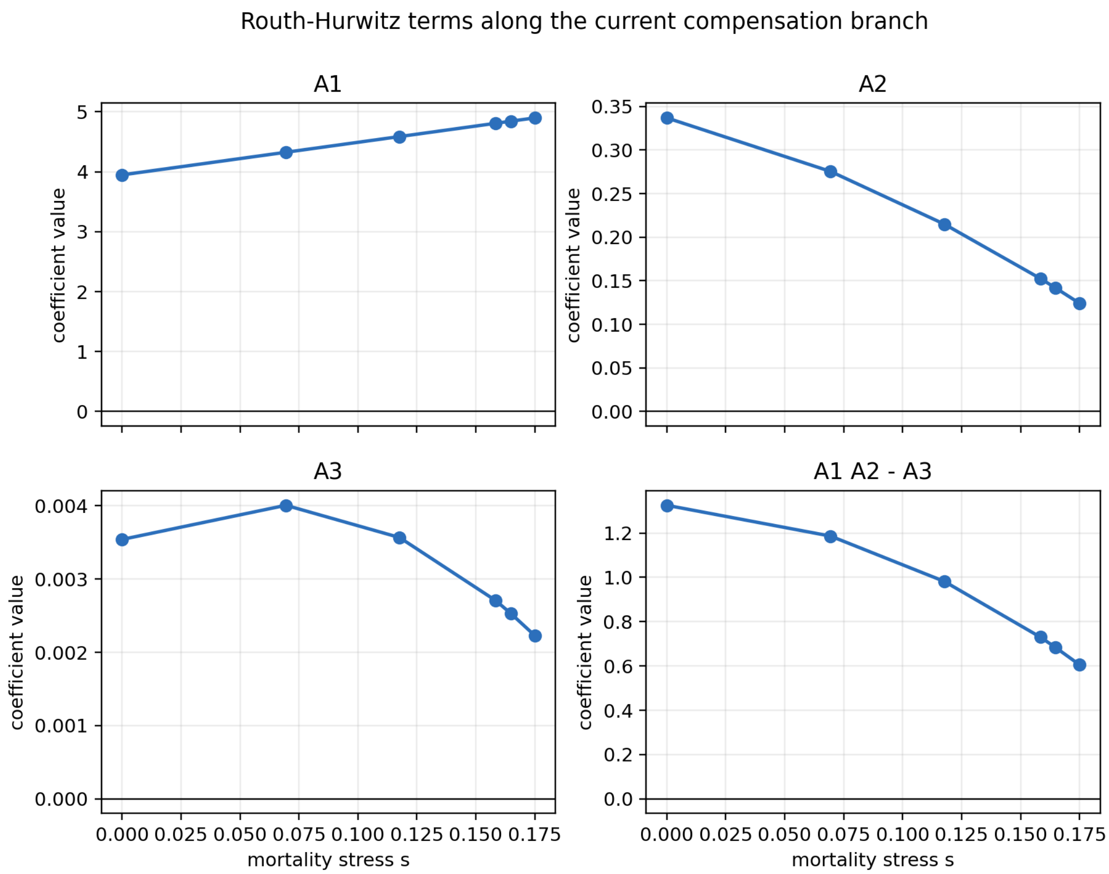

Figure 3: Routh–Hurwitz coefficient terms along the current compensation branch. Positivity of A_1, A_2, A_3, and $A_1A_2 - A_3$ supports local stability at the tested stresses.

6 Spatial PDE stability and non-homogeneous perturbations

A homogeneous ODE equilibrium is also a homogeneous PDE steady state because diffusion terms vanish for spatially constant fields. For Neumann spatial modes on a rectangular domain, perturbations evolve according to

$$J_F(U^*) - \lambda_{mn} D,$$

where $J_F(U^*)$ is the reaction Jacobian at the homogeneous branch state, $D = \mathrm{diag}(D_n, D_w, D_q)$, and

$$\lambda_{mn} = \left(\frac{m\pi}{L_x}\right)^2 + \left(\frac{n\pi}{L_y}\right)^2.$$

The zero mode is the ODE stability problem. Nonzero modes test whether spatial perturbations grow.

For the current diffusion coefficients, all tested nonzero Neumann modes were stable at the target branch stresses. A controlled diffusion-ratio linear grid also found no spatial instability in the tested ratios. Thus the PDE does not destabilize the homogeneous branch through the tested infinitesimal spatial modes.

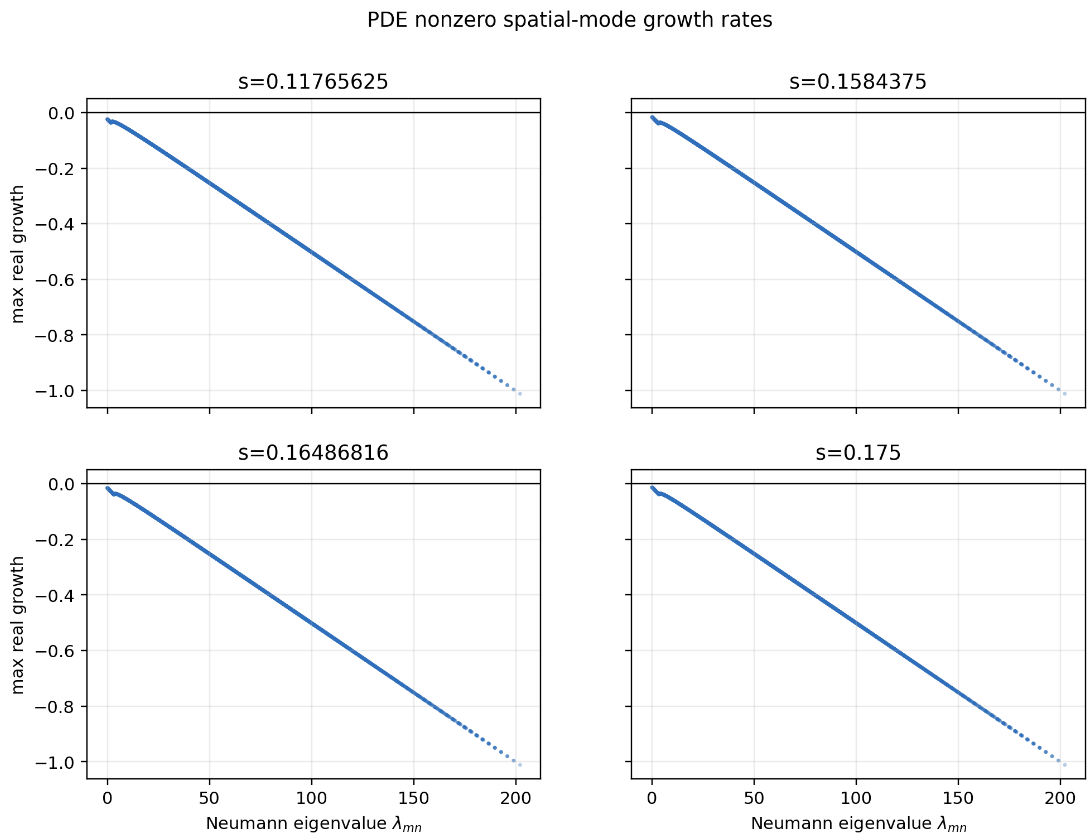

Figure 4: Maximum real growth rates for tested Neumann spatial modes around the homogeneous compensation branch. Nonzero modes have negative growth rates under the current diffusion coefficients.

Linear spatial stability is not enough to rule out finite-amplitude heterogeneous effects. Targeted nonlinear PDE tests therefore used localized predator patches, localized defense patches, sinusoidal perturbations, smooth random heterogeneity, and basin-boundary heterogeneity. These tests compared heterogeneous fields with matched homogeneous controls. Initial finite-horizon basin-label changes occurred for localized defense perturbations near basin-boundary mean states, but these changes were transient-label effects without persistent spatial patterning. Longer-horizon follow-up showed that the basin-changing cases resolved to the matched homogeneous-control basin, with final spatial coefficients of variation below the persistence threshold.

The spatial interpretation is therefore specific: in the tested PDE, spatial heterogeneity can create transient excursions near basin boundaries, but it did not generate a persistent spatial-pattern-mediated rescue mechanism.

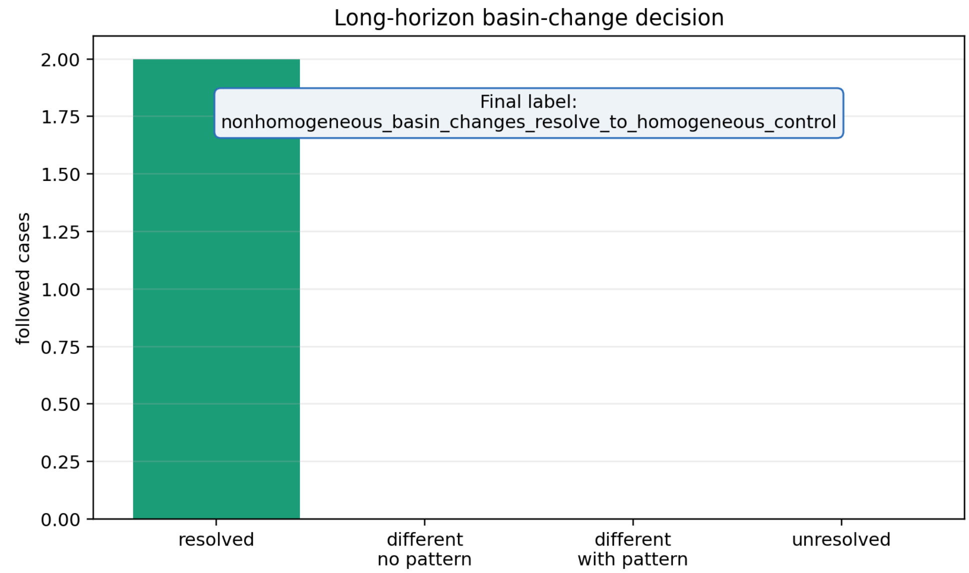

Figure 5: Long-horizon follow-up of the finite-amplitude non-homogeneous basin-changing cases. The followed cases resolved to their matched homogeneous-control basins and did not retain persistent spatial patterns.

7 Extension to nonlinear trade-off shapes

For nonlinear trade-offs, the branch is most naturally parameterized by q. When derivatives are defined and $a'(q) \neq 0$, the selection-gradient condition gives

$$c(q) = \frac{r'(q)}{a'(q)}.$$

The corresponding branch geometry is

$$z(q) = \frac{\xi}{r(q) - a(q)c(q)},$$

$$w(q) = c(q)z(q),$$

$$n(q) = \kappa^{-1} - z(q) - w(q),$$

and predator balance maps the branch to stress:

$$s(q) = b(q)n(q)z(q) - m - \mu w(q).$$

The linear branch is recovered when all shape exponents are one.

The controlled nonlinear extension used $\gamma_r, \gamma_a, \gamma_b \in \{0.5, 1, 2\}$, representing concave, linear, and convex endpoint-preserving shapes. This grid is small and structured; it is not a global trade-off scan. Within that grid, the generalized formulation recovered the linear branch, and selected concave, convex, and mixed nonlinear shapes supported feasible locally stable compensation branches. The shape-grid summary recorded 11 robust compensation shapes, 11 partial compensation shapes, 4 no-compensation shapes, and 1 unresolved shape. Targeted PDE spatial-mode and non-homogeneous tests for selected stable nonlinear branches did not support persistent spatial-pattern-mediated rescue.

The nonlinear result strengthens the mechanism by showing that compensation is not only a linear-trade-off artifact. It also emphasizes parameter sensitivity: nonlinear shape can change branch feasibility, stability, and stress interval length.

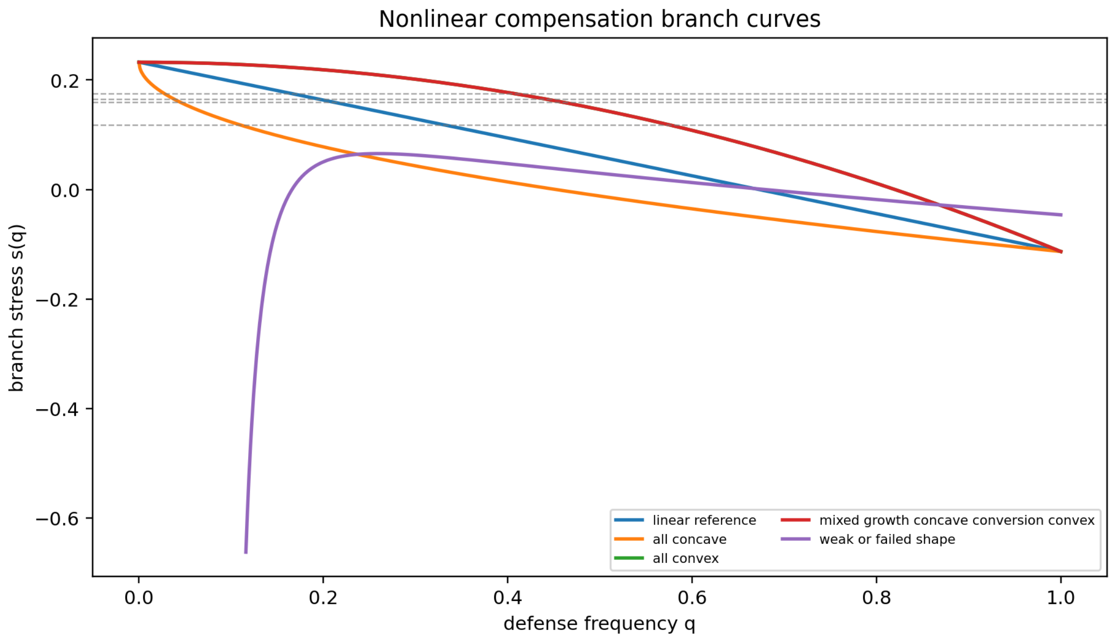

Figure 6: Selected nonlinear compensation branch curves. The linear case is recovered when all exponents equal one, while concave, convex, and mixed shapes can alter the stress range and q-location of feasible branch states.

8 Biological interpretation

Higher predator mortality directly reduces predator persistence. In this model, prey defense evolution can compensate because mortality stress reduces predator pressure, and lower predator pressure favors lower prey defense. Lower defense increases predator conversion opportunity through $b(q)$, which can restore predator growth balance at a positive predator equilibrium. This is indirect evolutionary rescue: the predator is rescued not by predator adaptation, but by prey evolution.

The spatial PDE does not create the rescue mechanism in the tested setup. Instead, the PDE preserves a homogeneous reaction-level branch that is locally stable to tested spatial modes. Spatial heterogeneity can matter transiently near basin boundaries, but the targeted tests did not produce persistent spatial patterning or spatial-pattern-mediated rescue.

9 Limitations

The linear compensation branch result is exact for the linear trade-off ODE under the stated feasibility conditions, but it is not automatically a theorem for all trade-off forms. The nonlinear trade-off extension is controlled and local, using only a structured exponent grid. The PDE tests are targeted rather than exhaustive. There is no global basin proof, no proof of global stability, no proof over all heterogeneous initial conditions, and no full bifurcation analysis.

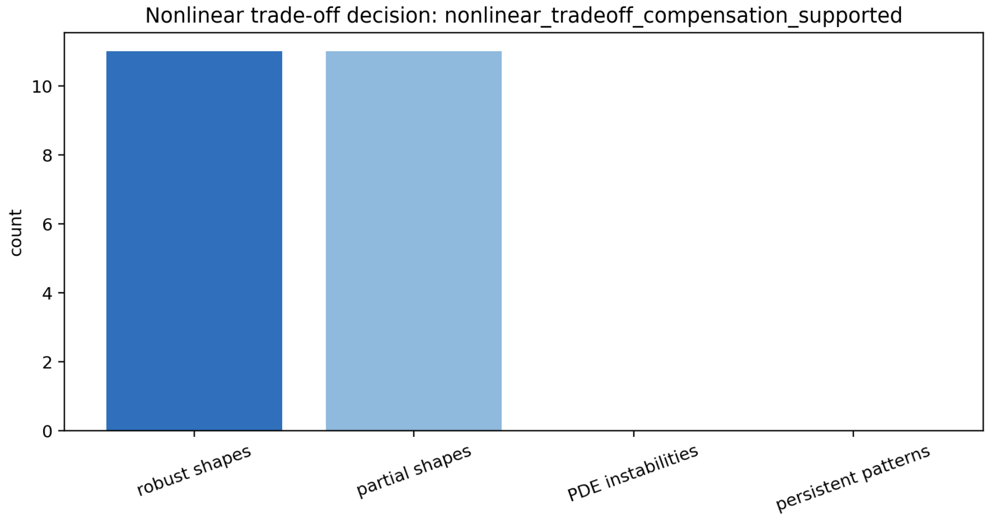

Figure 7: Summary of the controlled nonlinear trade-off extension. The final label supports nonlinear compensation in the selected structured grid while keeping the interpretation local and parameter-sensitive.

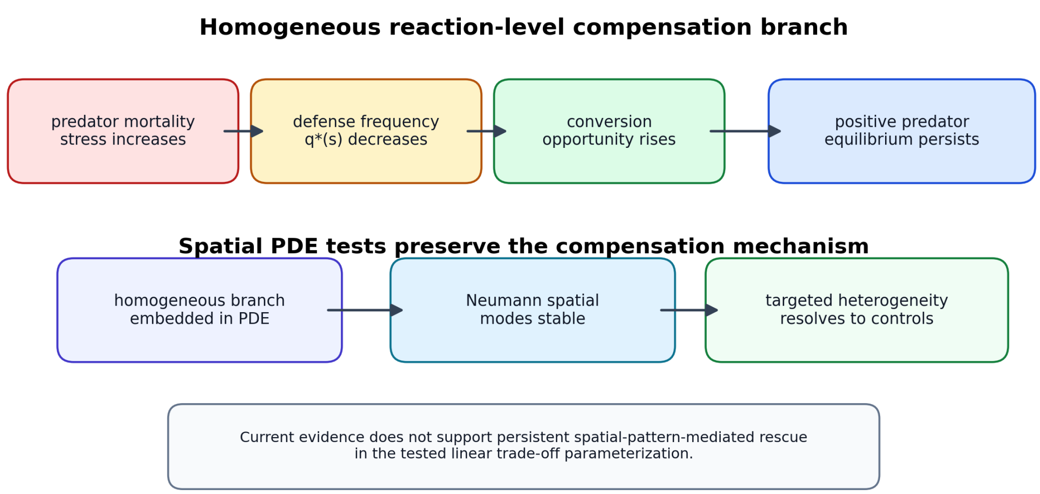

Figure 8: Final mechanism synthesis. Predator mortality stress shifts the homogeneous eco-evolutionary balance; lower prey defense compensates by increasing predator conversion opportunity. The spatial PDE preserves this reaction-level mechanism in the tested setup.

The model is not biologically calibrated. Parameters are used to test mechanism, not to fit an empirical predator–prey system. The results should therefore be interpreted as a mechanistic case study: they support homogeneous reaction-level compensation in the tested Roy-style model, not a universal claim about spatial evolutionary rescue.

10 Conclusion

1. The tested ODE supports indirect evolutionary rescue through prey defense evolution.
2. The mechanism is a homogeneous compensation branch.
3. Branch existence follows from explicit feasibility conditions.
4. The current branch is locally stable by Routh–Hurwitz criteria.
5. The spatial PDE branch is linearly stable to tested Neumann spatial modes.
6. Targeted non-homogeneous perturbations do not produce persistent spatial-pattern-mediated rescue.
7. Controlled nonlinear trade-off tests show that the mechanism extends beyond the linear case but remains parameter-sensitive.
8. The current result should be interpreted as homogeneous reaction-level compensation, not spatial-pattern-mediated rescue.

A Derivation of the compensation branch

At an interior equilibrium, the prey equation, predator equation, and selection-gradient equation are all zero after division by the positive factors n, w, and $\nu q(1-q)$. The selection-gradient equation gives

$$\Delta r z - \Delta a w = 0.$$

If $c = \Delta r/\Delta a > 0$, this yields $w = cz$. Substituting into prey balance,

$$r(q)z - \xi - a(q)cz = 0.$$

For linear trade-offs,

$$r(q) - ca(q) = r_u - ca_u,$$

so

$$z^* = \frac{\xi}{r_u - ca_u}, \quad w^* = cz^*, \quad n^* = \kappa^{-1} - z^* - w^*.$$

Finally, predator balance gives

$$b(q^*) = \frac{m + s + \mu w^*}{n^* z^*},$$

and because $b(q) = b_u + \Delta b q$,

$$q^*(s) = \frac{(m + s + \mu w^*)/(n^* z^*) - b_u}{\Delta b}.$$

B Jacobian and Routh–Hurwitz coefficients

Write

$$F_n = nR, \quad R = r(q)z - \xi - a(q)w,$$
$$F_w = wP, \quad P = b(q)nz - (m+s) - \mu w,$$
$$F_q = \nu q(1-q)G, \quad G = \Delta rz - \Delta aw.$$

At the branch, $R = P = G = 0$. The analytic Jacobian was evaluated at the branch and summarized by the characteristic polynomial

$$\det(\lambda I - J) = \lambda^3 + A_1\lambda^2 + A_2\lambda + A_3.$$

The coefficients were computed as

$$A_1 = -\mathrm{tr}(J),$$
$$A_2 = \frac{1}{2}\left[(\mathrm{tr}J)^2 - \mathrm{tr}(J^2)\right],$$
$$A_3 = -\det(J).$$

The Routh–Hurwitz stability inequalities are $A_1 > 0$, $A_2 > 0$, $A_3 > 0$, and $A_1A_2 > A_3$.

C Numerical protocols and classification rules

The manuscript consolidates existing outputs. It does not introduce new simulations or parameter scans. ODE/PDE comparisons used matched initial means. Spatial-mode tests used Neumann eigenvalues and the modal matrix $J_F(U^*) - \lambda_{mn}D$. Non-homogeneous PDE tests compared each heterogeneous run with a matched homogeneous control at the same stress and baseline state. Basin-label changes were interpreted only after long-horizon follow-up when available. Persistent spatial patterning was assessed with final spatial coefficient-of-variation thresholds from the corresponding research notes.

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